BTS-VII-11.17-1230 Reg.No. | | | | | | | | |

B

B.Tech. Degree VII Semester Examination November 2017

ME 1701 REFRIGERATION AND AIR CONDITIONING

(2012 Scheme)

Time : 3 Hours Maximum Marks : 100

PART A

(Answer *ALL* questions)

($8 \times 5 = 40$)

I. (a) Define one tonne of refrigeration.

(b) Describe briefly the working of Pulse tube refrigeration with a neat figure.

(c) In a vapour absorption refrigeration, heat is supplied to the generator at a temperature of 90°C. The cooling in condenser and evaporator takes place at 20°C and -10°C respectively. Find the maximum COP of the system.

(d) List the differences between vapour compression and vapour absorption refrigeration system.

(e) Explain the working of an axial flow compressor. How does it differ from a centrifugal compressor?

(f) Describe briefly the vane type compressor with a neat sketch.

(g) Write short notes on by-pass factor for cooling coils.

(h) What is 'effective temperature'? What are the factors governing effective temperature?

PART B

($4 \times 15 = 60$)

II. (a) Ice is formed at 0°C from water at 20°C. The temperature of the brine is -8°C. (8)
Find (i) COP (ii) kilogram of ice formed per kWh. Assume the cycle is reversed Carnot cycle. Take latent heat of ice as 335 kJ/kg.

(b) In a Bell-Coleman cycle air refrigerator, air is compressed to 5 bar from 1 bar and its initial temperature is 10°C. After compression, the air is cooled upto 20°C in a cooler. Calculate COP and net refrigerating effect. Take $Cp = 1.005\ kJ/kgK$ and $Cv = 0.718\ kJ/kgK$. (7)

OR

III. (a) What is meant by thermo electric refrigeration? Explain. (5)

(b) With the help of p-v and T-s diagrams, explain the working of Bell-Coleman cycle air refrigerator. (10)

(P.T.O.)

IV. A Freon – 12 vapour compression refrigeration system has a condensing temperature of 50°C and evaporating temperature of 0°C. The refrigeration capacity is 7 tonnes. The liquid leaving the condenser is saturated liquid and compression is isentropic. Determine : (15)

(i) Mass flow rate
(ii) Power required to run the compressor.
(iii) COP of the system.

The properties of Freon – 12 are

Temperature (°C)	Pressure (bar)	Enthalpy (kJ/kg)		Enthalpy (kJ/kgK)	
		Liquid h_f	Vapour h_g	Liquid S_f	Vapour S_g
50°C	12.199	84.868	206.98	0.3034	0.6792
0°C	3.086	36.022	187.397	0.1418	0.6960

OR

V. (a) Briefly explain the practical ammonia-water absorption refrigeration system with a neat figure. (9)

(b) Write short note on multi pressure refrigeration systems. (6)

VI. (a) In a two stage reciprocating compressor, derive an expression of intercooler pressure (p_2) as the geometric mean of suction pressure (p_1) and condenser pressure (p_3). (9)

(b) Define volumetric efficiency of a reciprocating compressor and also derive an expression for volumetric efficiency. (6)

OR

VII. Explain briefly the types of (i) condensers and (ii) evaporators used in refrigeration systems with neat sketches. (15)

VIII. (a) Differentiate between summer and winter air conditioning systems. (5)

(b) In a laboratory test, a psychrometer recorded dry bulb temperature as 35°C and wet bulb temperature as 28 °C. Calculate : (10)

(i) Vapour pressure.
(ii) Relative humidity.
(iii) Specific humidity.
(iv) Dew point temperature.
(v) Enthalpy.

OR

IX. (a) What are the procedures for cooling load calculation of an air conditioned system? (5)

(b) Explain duct design methods and air distribution arrangements in air conditioning systems. (10)

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